

Simran Bilkhu

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SUMMARY

M.Sc. Statistics, McMaster University (2026), with thesis-level research in Bayesian factor models and a co-authored publication in probability theory. Applies rigorous statistical methods to real problems: prior work spans spatio-temporal modelling, high-dimensional classification, and causal inference; contributed to over \$50,000 in cost reductions at Sodexo Canada using data analytics.

EDUCATION

• McMaster University

M.Sc. Statistics (Thesis) — GPA 11.86/12

◦ Thesis [\[link\]](#)

A Comparative Study of Bayesian Factor Models under PLT and UGLT Identification Constraints

September 2024 – April 2026

Hamilton, Canada

• University of Toronto

H.B.Sc. Applied Statistics, Minor in Mathematics (Honours) — GPA 3.76/4.00 (final 2 yrs)

April 2024

Toronto, Canada

WORK EXPERIENCE

• McMaster University

Research Assistant II

May 2026 – Present

Hamilton, ON

- Continuing research in Bayesian factor models post-graduation; conducting simulation studies in **R** and contributing to methodology development and academic writing.

• McMaster University

Teaching Assistant — Statistical Methods, Linear Regression Using SAS, Intermediate Probability Theory

Sep. 2024 – Dec. 2025

Hamilton, ON

- Delivered tutorials and office hours for undergraduate statistics courses (Statistical Methods, Intermediate Probability Theory); graded 500+ assessments with written feedback.
- Conducted live **SAS** lab tutorials for Linear Regression Using SAS; provided hands-on coding support and constructive feedback on analysis reports.

• Sodexo Canada

Intern

March 2020 – August 2022

Oakville, ON

- Analyzed telecom contract and operational cost data to identify savings opportunities, contributing to over **\$50,000** in cost reductions; built dashboards to communicate findings and project progress to senior leadership.
- Led Canada-wide tracking and replacement of point-of-sale terminals; managed structured data across hardware asset management and operational cost-reduction initiatives.

PROJECTS

• Bird Migration Count Forecasting

R, Poisson Regression, Gaussian Random Fields

December 2024

[Link](#)

- Forecasted population trends of *Peregrine Falcons* and *Greater Roadrunners* using satellite climate data: modelled counts (13,642 obs, 2001–2005) from time, location, and weather features; a spatio-temporal mixed model **reduced test error from 1,377 to 31** vs. ridge-penalised Poisson, with warmer temperatures positively associated with counts.

• Baseline Climate Partitions via Clustering

Python, scikit-learn

November 2024

[Link](#)

- Defined stable baseline climate regions across North America: benchmarked K-means and agglomerative clustering on **~130K observations** across **19 bioclimatic features**; identified 5 distinct climatological clusters.

• Diet Intervention and Gut Microbiome

R, PCA, GMM, Random Forest

January – April 2025

[Link](#)

- Investigated how diet shapes gut microbial composition: preprocessed a $236 \times 1,415$ -taxon matrix, reduced via PCA (20 components, >60% variance), clustered with GMM, and trained a Random Forest classifier (4,000 trees) achieving **91% accuracy** (adjusted Rand index 0.68).

TECHNICAL SKILLS

- **Programming languages & Tools:** Python (NumPy, Pandas, scikit-learn, Matplotlib), R (tidyverse, Stan, brms, lme4, Shiny), SQL, SAS, LaTeX, Git

- **Statistics & ML:** Bayesian and frequentist inference, causal inference (experimental design, A/B testing, instrumental variables, SEMs), GLMs & GLMMs, survival analysis, time series, PCA, clustering (K-means, GMM, agglomerative), random forests, gradient boosting, neural networks, regularization (Lasso, Ridge, elastic net).

- **Platforms:** Jupyter, RStudio, VS Code, Shiny, Quarto, GitHub, Linux

PUBLICATIONS & PRESENTATIONS

Paper: *Counter-Examples to Continuity of Measure in Uncountable Unions* (with N. M. Forman) [\[link\]](#).

Talks: Sparse Bayesian Factor Analysis [\[slides\]](#), Sparse PCA [\[slides\]](#), HMC [\[slides\]](#), Bayesian Workflow